



FASR Science Case # 14 (local draft)

Full-Disk Microwave Imaging of Active-Region Coronal Magnetism and the Sources of F10.7

Sijie Yu¹, and FASR team

¹Center for Solar-Terrestrial Research, New Jersey Institute of Technology

1 Science Goal(s)

Briefly summarize the key science goal(s) for this science case. A few sentences will be sufficient.

This science case addresses two linked questions. First, how do the strong coronal magnetic fields above active regions change as magnetic flux emerges, evolves, and approaches eruption? Second, what fractions of the disk-integrated F10.7 flux arise from thermal free-free emission and from magnetic-field-dependent gyroresonance emission? FASR will answer these questions with repeated full-disk Stokes I and V imaging spectroscopy, combined with photospheric magnetic fields, EUV thermal constraints, and gyroresonance-plus-free-free radiative-transfer modeling. The resulting record will follow active-region coronal field-strength layers over hours to days and give F10.7 a spatially resolved physical interpretation. For space weather, the field maps add a time-dependent constraint on active-region eruptive state. The separated free-free component can also be tested as a solar-EUV proxy for ionospheric, thermospheric, satellite-drag, and radio-propagation applications.

2 Scientific Rationale

2.1 Scientific Importance

Provide a brief discussion on the scientific importance of this science case.

Active regions store the magnetic free energy that powers major flares and coronal mass ejections, yet their coronal magnetic fields are not measured routinely across the visible disk. Photospheric magnetograms constrain the lower boundary, while EUV and X-ray images primarily trace plasma shaped by the field. Gyroresonance supplies a complementary constraint on the field itself. Emission becomes optically thick near low harmonics of the electron gyrofrequency,

$$\nu_{\text{GHz}} \simeq 2.8 \times 10^{-3} s B[\text{G}],$$

so the frequency at which a source changes opacity identifies candidate iso-Gauss layers when the harmonic s is established. At 2.8 GHz, harmonics $s = 1\text{--}4$ correspond to approximately 1000, 500, 333, and 250 G. Broadband morphology, spectral cutoffs, and Stokes V can therefore constrain the location, shape, and field strength of the strong coronal structures above sunspots [Gary 2023](#).

Repeating the measurement over hours and through disk passage will show whether those structures strengthen, expand, move, or decay as an active region evolves. These measurements are relevant to space weather because the coronal field contains the free-energy reservoir and overlying confinement that govern whether an active region can produce a major eruption. Synoptic FASR products would provide time-dependent constraints for magnetic and eruption models; they should not be presented as a stand-alone flare or CME forecast until prospective predictive skill is demonstrated. HMI boundary fields, AIA thermal constraints, and exact gyroresonance-plus-free-free transfer are required to separate intrinsic magnetic evolution from temperature, opacity, harmonic assignment, and viewing geometry; a radio frequency is not by itself a geometric height.

F10.7 is the disk-integrated solar flux at 2.8 GHz. Its stable record since 1947 makes it a principal proxy for the solar EUV input used in ionospheric and thermospheric modeling [Tapping 2013](#). Those models support radio-propagation and GNSS applications and estimates of thermospheric neutral density; neutral-density errors propagate into satellite-drag and conjunction calculations [Bilitza et al. 2017](#); [Emmert 2015](#). The physical meaning of the index is nevertheless ambiguous because it combines optically thin free-free emission from thermal plasma with optically thick gyroresonance emission above strong active-region fields. Free-free emission is closely related to the plasma producing EUV radiation; gyroresonance is weighted toward strong magnetic fields and can grow disproportionately at high activity. The F10.7–EUV relation consequently becomes nonlinear when the Sun is active [Schonfeld et al. 2019](#).

Full-disk imaging spectroscopy makes this mixture measurable. A calibrated 2.8 GHz map locates the flux, neighboring frequencies identify the local spectral form, and Stokes V and magnetic context help distinguish gyroresonance from free-free emission. FASR can partition the total flux among active regions, plage, the quiet Sun, and coronal holes, then integrate the modeled components to measure the active-region gyroresonance contribution to F10.7. Earlier single-day imaging showed the power of this approach [Schonfeld et al. 2015](#); FASR can turn it into a routine record spanning rotation and the solar cycle. The result is a physically interpretable image-derived F10.7 product that can be tested directly against EUV irradiance rather than corrected only through an empirical running average. If the free-free component improves EUV and upper-atmosphere reconstructions, it would provide a new ground-based space-weather product whose physical source terms and uncertainty can be inspected on the solar disk.

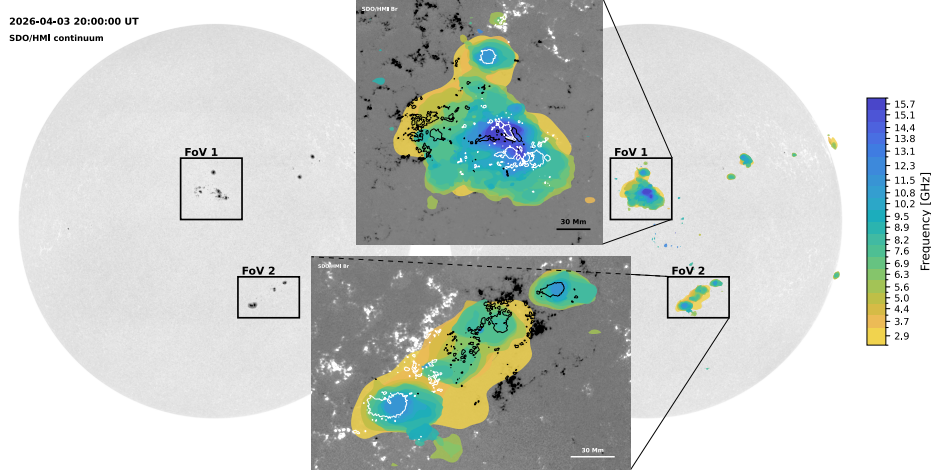


Figure 1: EOVSA active-region footprints at 2.9–15.7 GHz over HMI continuum and radial-field maps. Their contraction with increasing frequency illustrates how broadband source boundaries sample progressively stronger coronal fields. The colors mark frequency, not intensity. Adapted from Yu et al. (2026).

2.2 Uniqueness to FASR Capabilities

Is this science case uniquely addressed by FASR? Why can't other facilities address this science and achieve the same goal?

Existing solar arrays establish the emission physics but do not provide routine, full-disk imaging spectroscopy with the spatial, spectral, temporal, and polarimetric information required here. FASR-A's 120 antennas provide 7140 instantaneous baselines. The science mode is therefore a sequence of 10 s full-disk snapshots, each containing calibrated Stokes I and V across the magnetic-diagnostic band. This cadence separates true field evolution from solar rotation and allows the same data stream to follow flux emergence and rapid pre-eruptive changes.

The frequency tuning must sample gyroresonance opacity cutoffs closely enough to locate their source boundaries. The two circular polarizations can reveal different adjacent harmonics. FASR must measure calibrated total power throughout 1–20 GHz, alongside the interferometric data. It anchors absolute brightness and disk-integrated flux across free-free, gyroresonance, and gyrosynchrotron spectra; the 2.8 GHz measurement also supplies F10.7. Combining total power with resolved images requires short-spacing coverage and stable radiometric calibration. EOVSAs are the pathfinder for the analysis and calibration methods. Its long Earth-rotation synthesis is specific to that array.

2.3 Synergies

Describe potential synergies/complementarities between this FASR science case and those from current/future/planned facilities at all wavelengths (e.g., DKIST, MUSE, FIERCE, COSMO, ngGONG, etc.).

SDO/HMI provides the photospheric magnetic boundary and continuum context, while SDO/AIA supplies the differential-emission-measure and morphological constraints needed to model free-free and gyroresonance opacity. Direct EUV irradiance measurements from SDO/EVE, GOES/EXIS, and

related records provide the decisive test of whether the image-derived, gyroresonance-corrected F10.7 component is a better physical EUV proxy. Comparison with ionospheric and thermospheric models will then test whether that improvement is material for operationally relevant electron-density and neutral-density estimates. DKIST and future coronal spectropolarimeters can test selected magnetic inferences at the limb or in targeted fields. MUSE and Solar-C/EUVST can add temperature, density, and velocity information for active regions. The operational F10.7 record and RSTN measurements provide long-term radiometric comparisons, but they are external checks rather than substitutes for an independent FASR flux and brightness calibration.

2.4 Measurements Required by FASR

Provide a description of the necessary measurements to be carried out by the FASR to adequately address this science case. Please coordinate these measurements with the Science Requirements table in Section III.

FASR should image the entire solar disk and several arcminutes beyond the limb from 1 to 20 GHz. The baseline product is a calibrated Stokes I and V cube every 10 s with contemporaneous total-power measurements at every sampled frequency. The band covers low-harmonic gyroresonance from roughly 200 G to more than 2000 G and includes 2.8 GHz for F10.7. Frequency samples must be close enough that their contribution to the field-strength uncertainty is at most 2 percent. Since $B \propto \nu/s$, this requires $\Delta\nu/\nu \leq 0.02$, or about 150 independent samples over 1–20 GHz. The tuning sequence must complete within the 10 s cadence. Q and U are not required because coronal Faraday rotation, unresolved structure, and bandwidth and beam averaging strongly suppress linear polarization at these wavelengths.

For each well-measured active region, the analysis should deliver candidate iso-Gauss surfaces, thermal and opacity maps conditional on the HMI-constrained field, and a field-strength time series supported by Stokes I and V. Across the sampled band, the transfer model should separate free-free and gyroresonance brightness in each well-measured image pixel and test for any gyrosynchrotron contribution. Integrating the 2.8 GHz component maps provides the modeled total F10.7 flux, the active-region gyroresonance fraction, and an image-derived free-free component to compare with simultaneous EUV irradiance. These products require matched-time HMI and AIA inputs and must carry the radio calibration, imaging, thermal-model, and harmonic-assignment uncertainties.

The measurement succeeds when the snapshot spectra recover the gyroresonance cutoff, the correct adjacent-harmonic pair, and the free-free/gyroresonance partition. White (2004) shows observed active-region brightness up to about 5×10^6 K and useful source-boundary contours near 5×10^4 K. Detecting that boundary at five standard deviations gives a science-derived snapshot-image rms requirement of 10^4 K per spectral sample in both I and V, together with an artifact-limited dynamic range of at least 500:1. These values replace the earlier hour-integrated 5000 K assumption. End-to-end tests must confirm that they deliver magnetic-field strength and active-region component fluxes to 10 percent, with 5 percent as a goal. Total-power measurements are mandatory across the observed band: an interferometer does not measure the zero-spacing visibility, so interferometric images alone cannot set the absolute brightness and disk-integrated flux scale for free-free, gyroresonance, or gyrosynchrotron emission. At 2.8 GHz, this also supplies the absolute F10.7 flux [Thompson et al. 2017](#).

III. Science Requirements Tables

(A) Observational Target Description

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

The target is the full radio Sun rather than a collection of separately pointed sources. A $40' \times 40'$ field contains the approximately $32'$ disk and provides off-limb context. Thermal active-region gyroresonance can reach several million kelvin, while the scientifically useful opacity boundary falls to chromospheric brightness. The measurement must retain that boundary throughout disk passage and during changes in viewing angle.

(A) OBSERVATIONAL TARGET		
Type of observation (what defines a ‘target’)	One full-Sun microwave image cube containing the quiet disk, coronal holes, active regions, and off-limb emission.	
Number of targets	One Sun per observing day; all on-disk regions are measured simultaneously.	
Size of a single target (arcsec x arcsec)	At least $2400'' \times 2400''$, including off-limb context around the approximately $1920''$ optical disk.	
Distribution of all targets (arcmin x arcmin)	One $40' \times 40'$ field centered on the Sun.	
Peak brightness (sfu/beam or Kelvin)	Frequency and activity dependent; approximately 10^5 K for much of the disk and up to about 5×10^6 K in bright thermal active-region gyroresonance sources.	
RMS brightness (sfu/beam or Kelvin)	Required snapshot-image rms: $\leq 10^4$ K in Stokes I and V per magnetic-diagnostic spectral sample in a complete 10 s cube.	
Expected circularly polarized flux density	Stokes V (sfu/beam)	Source dependent. Calibrated V is a baseline measurement because the two modes can show different harmonic cutoffs.
	V/I	Near zero over much of the quiet disk. In gyroresonance sources, the sign and magnitude depend on harmonic, field polarity, viewing angle, and transfer physics; no fixed full-disk fraction is assumed.
Expected linearly polarized flux density	Stokes Q or U (sfu/beam)	Not required.
	Q/I or U/I	Not required; coronal propagation and unresolved structure strongly depolarize Q and U at centimeter wavelengths.

(B) Spectro-Temporal Requirements Discussion

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

The 1–20 GHz range samples low-harmonic gyroresonance across the active-region field strengths of interest. A sharp spectral cutoff identifies an iso-Gauss boundary; two cutoffs in opposite circular polarizations can identify adjacent harmonics through their frequency ratio [White 2004](#). A 2 percent fractional spacing limits frequency-sampling error in B to 2 percent within a fixed harmonic and yields about 150 samples over the band. The full Stokes I/V sequence and corresponding total-power samples must be acquired within 10 s so their spectra describe the same solar state.

(B) SPECTRAL-TEMPORAL REQUIREMENTS	
Central Frequency (GHz)	N/A; broadband 1–20 GHz, including a calibrated 2.8 GHz layer
Instantaneous Bandwidth (GHz/pol)	Frequency-agile coverage sufficient to complete 1–20 GHz Stokes I/V sampling within 10 s
Spectral resolution [MHz]	$\Delta\nu/\nu \leq 2\%$; use finer sampling where a cutoff would otherwise be unresolved
Temporal resolution (in seconds)	10 s for a complete full-band Stokes I/V image cube

(C) Polarization Product Discussion

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

Stokes I and calibrated Stokes V are both baseline products. The extraordinary and ordinary modes can become optically thick at different harmonics, so either polarization can contain the cutoff needed to determine the field. Q and U do not contribute to this retrieval and are strongly reduced by coronal propagation and spatial averaging.

(C) POLARIZATION DATA PRODUCTS REQUIRED	
y	Stokes I (baseline)
n	Stokes Q
n	Stokes U
y	Stokes V (baseline)

(D) Imaging Requirements Discussion

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

The image must recover scales from compact active-region structure to the full $2000''$ -scale disk. The active-region requirement is $\leq 10''$ resolution at 5–10 GHz, sufficient to separate source boundaries across sunspot-scale magnetic gradients; pixels should sample the synthesized response at least three times. Spectral inference uses the measured frequency-dependent response directly in forward modeling.

The 10^4 K snapshot rms follows from detecting a 5×10^4 K source boundary at five standard deviations. The 500:1 image dynamic range follows from retaining that boundary in the presence of active-region peaks approaching 5×10^6 K. These requirements apply to I and V in the spectral samples used for the magnetic retrieval. The absolute flux scale across 1–20 GHz must be accurate to 5 percent, with 1–3 percent as the goal, so the spectral inversions and F10.7 component partition are not calibration limited.

(D) IMAGING REQUIREMENTS	
Required angular resolution (arcsec) (single value or range)	$\leq 10''$ at 5–10 GHz for active-region magnetometry; use the measured frequency-dependent response in the inversion.
Largest angular scale required (arcsec)	At least $2000''$, covering the complete radio disk.
Mapped image size (arcmin x arcmin)	$40' \times 40'$.
Required pixel resolution (arcsec)	No larger than one third of the synthesized response; $\leq 3''$ pixels at 5–10 GHz.
Number of output/image channels	At least 150 independently calibrated continuum samples across 1–20 GHz, including 2.8 GHz and finer sampling at opacity cutoffs.
Output bandwidth (minimum and maximum frequency - GHz)	1–20 GHz baseline coverage.
Channel width (MHz)	Frequency dependent, with $\Delta\nu/\nu \leq 2\%$.
Required rms (sfu/beam or Kelvin) [per channel] (if polarization products required define for each)	Stokes I and V: $\leq 10^4$ K per magnetic-diagnostic spectral sample in a 10 s snapshot.
Dynamic range within image (if polarization products required define for each)	At least 500:1 artifact-limited dynamic range in 10 s Stokes I and V snapshots; 1000:1 goal.
Polarization accuracy (%)	I-to-V leakage $\leq 1\%$ of local I; 0.3% goal.

(D) IMAGING REQUIREMENTS (continued)		
Zero spacing/total power required?	Yes, at every sampled frequency across 1–20 GHz and contemporaneous with imaging. Total power anchors absolute brightness and integrated flux for free–free, gyroresonance, and gyrosynchrotron emission; 2.8 GHz also supplies F10.7.	
Required maximum latency (in seconds, or N/A)	60 s for a quicklook 10 s cube; final calibrated products may follow later.	
Required flux density scale calibration accuracy	G	1-3%
	B	5%
		10%
		20-50%

2.5 Other Performance or Functional Requirements

If there are any additional performance or functional requirements not captured above, describe them here. For example, beamforming array mode, phased array, etc.

Each 10 s cube must carry calibration, image-quality, and uncertainty information, including contemporaneous total-power spectra across the sampled band. The pipeline should publish I and V image cubes, gyroresonance cutoff and harmonic-assignment maps, candidate iso-Gauss maps, and active-region field-strength time series. Across the sampled band it should publish calibrated total-power spectra and free–free, gyroresonance, and, where detected, gyrosynchrotron component spectra and maps. At 2.8 GHz it should also publish component disk-integrated fluxes and the resulting gyroresonance fraction of F10.7. Comparisons with simultaneous EUV irradiance test the space-weather value of the free–free component. The frequency-dependent instrumental response belongs in the forward model; no daily integration or common restoring beam is required to form the baseline product.

3 Appendix

Please provide any other relevant material necessary to understand and substantiate this Science Case.

EOVSA is the algorithm and calibration pathfinder for this case. Its data can be used to prototype full-disk brightness calibration, gyroresonance-plus-free–free transfer, and the F10.7 component inversion. Its long-track signal-to-noise gains cannot be scaled directly to FASR because the arrays have different instantaneous baseline coverage.

The FASR feasibility demonstration should inject an evolving, frequency-dependent full-Sun I/V model into the 120-antenna response and reconstruct independent 10 s snapshots with matching total-power spectra. The model must include a diffuse disk, coronal holes, limb structure, active-region free–free emission, polarized gyroresonance cutoffs, and representative gyrosynchrotron emission. After realistic gain, leakage, deconvolution, total-power calibration, and short-spacing errors are added, the test must recover the correct adjacent harmonics and field-strength layers, active-region component fluxes within 10 percent, broadband absolute fluxes within the specified calibration

budget, and the disk-integrated F10.7 partition within 10 percent. A second test should run the same inversion on EOVSa observations as an empirical check of the emission model. This establishes a path from a working prototype to FASR without importing EOVSa’s long-integration observing strategy.

Reference

- Gary, D. E., Chen, B., Dennis, B. R., et al. 2018, *ApJ*, 863, 83, “Microwave and Hard X-Ray Observations of the 2017 September 10 Solar Limb Flare.” <https://doi.org/10.3847/1538-4357/aad0ef>.
- Gary, D. E. 2023, *ARA&A*, 61, 427–472, “New Insights from Imaging Spectroscopy of Solar Radio Emission.” <https://doi.org/10.1146/annurev-astro-071221-052744>.
- Nakajima, H., Nishio, M., Enome, S., et al. 1994, *Proc. IEEE*, 82, 705–713, “The Nobeyama Radioheliograph.” <https://doi.org/10.1109/5.284737>.
- Shimojo, M., and Iwai, K. 2023, *Geoscience Data Journal*, 10, 114–129, “Over seven decades of solar microwave data obtained with Toyokawa and Nobeyama Radio Polarimeters.” <https://doi.org/10.1002/gdj3.165>.
- Yu, S., et al. 2026, “Synoptic Solar Microwave Observations from EOVSa. I: Rotation-aware Full-disk Imaging,” manuscript in preparation.
- White, S. M. 2004, “Coronal Magnetic Field Measurements through Gyroresonance Emission,” in *Solar and Space Weather Radiophysics: Current Status and Future Developments*, ed. D. E. Gary and C. U. Keller (Dordrecht: Kluwer), 89–112.
- Thompson, A. R., Moran, J. M., and Swenson, G. W. 2017, *Interferometry and Synthesis in Radio Astronomy*, 3rd ed. (Cham: Springer). <https://doi.org/10.1007/978-3-319-44431-4>.
- Schonfeld, S. J., White, S. M., Henney, C. J., Arge, C. N., and McAteer, R. T. J. 2015, *ApJ*, 808, 29, “Coronal Sources of the Solar F10.7 Radio Flux.” <https://doi.org/10.1088/0004-637X/808/1/29>.
- Schonfeld, S. J., White, S. M., Henney, C. J., Hock-Mysliwiec, R. A., and McAteer, R. T. J. 2019, *ApJ*, 884, 141, “The Slowly Varying Corona. II. The Components of F10.7 and Their Use in EUV Proxies.” <https://doi.org/10.3847/1538-4357/ab3af9>.
- Tapping, K. F. 2013, *Space Weather*, 11, 394–406, “The 10.7 cm Solar Radio Flux (F10.7).” <https://doi.org/10.1002/swe.20064>.
- Bilitza, D., Altadill, D., Truhlik, V., et al. 2017, *Space Weather*, 15, 418–429, “International Reference Ionosphere 2016: From Ionospheric Climate to Real-Time Weather Predictions.” <https://doi.org/10.1002/2016SW001593>.
- Emmert, J. T. 2015, *Advances in Space Research*, 56, 773–824, “Thermospheric Mass Density: A Review.” <https://doi.org/10.1016/j.asr.2015.05.038>.